

Semantic Consistency Outperforms Token Entropy for Hallucination Detection: A Systematic Comparison

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Abstract

Large language models generate plausible but factually incorrect outputs, motivating uncertainty-based hallucination detection. Token entropy and semantic consistency are two promising signals, yet their relative effectiveness and potential complementarity remain unstudied. We present the first systematic comparison of these signals on TriviaQA using Llama-2-7B-chat. Our experiments reveal that semantic consistency—measured via pairwise embedding similarity across multiple samples—achieves AUROC 0.81 with large effect size (Cohen’s $d=1.19$), dramatically outperforming token entropy (AUROC 0.65, $d=0.47$). Surprisingly, linear fusion of these signals yields no improvement over consistency alone at proof-of-concept scale ($N=20$), with optimal weights collapsing to pure consistency. The moderate negative correlation between signals ($r=-0.54$) suggests partial redundancy rather than orthogonality. Our findings indicate practitioners should prioritize consistency-based detection for factual QA; the complementarity hypothesis requires larger-scale validation before practical deployment.

1 Introduction

Large language models frequently generate fluent but factually incorrect responses—a phenomenon known as hallucination that undermines their reliability in knowledge-intensive applications. When an LLM confidently states that “Einstein won the 1922 Nobel Prize in Chemistry” rather than Physics, users have no way of knowing the response is wrong without external verification. This problem has motivated extensive research into uncertainty quantification: if we could reliably predict *when* models are likely to hallucinate, we could warn users or abstain from responding entirely.

Two families of uncertainty signals have emerged as promising hallucination detectors. **Token-level entropy**, computed from the model’s softmax distribution over next tokens, captures the model’s internal uncertainty—when logit distributions are flat (high entropy), the model has low confidence in any particular continuation [Kuhn et al., 2023]. **Semantic consistency**, measured by generating multiple responses and computing their pairwise similarity, captures output stability—when the model “knows” an answer, it produces consistent outputs; when hallucinating, outputs diverge [Manakul et al., 2023]. A natural hypothesis follows: since these signals capture different failure modes (internal confusion versus output instability), combining them should improve detection.

We test this hypothesis systematically—and find a counterintuitive result. On TriviaQA with Llama-2-7B-chat, semantic consistency alone achieves AUROC=0.81 with a large effect size (Cohen’s $d=1.19$), dramatically outperforming token entropy (AUROC=0.65, $d=0.47$). More surprisingly, linear fusion of the two signals provides *no improvement* over consistency alone at proof-of-concept scale ($N=20$). The signals are moderately negatively correlated ($r=-0.54$), suggesting partial redundancy rather than the complementarity we hypothesized.

1.1 The Problem at Three Levels

Surface problem: LLM hallucinations erode trust and cause harm in high-stakes applications including medical question answering, legal document analysis, and educational tutoring. Users cannot distinguish confident-but-wrong outputs from reliable ones.

Deeper problem: While multiple uncertainty signals exist, they have been studied in isolation. Kuhn et al. [2023] demonstrated that semantic entropy—entropy computed after clustering responses by meaning—improves calibration. Manakul et al. [2023] showed that self-consistency checking detects factual errors.

Xiong et al. [2023] found that models’ verbalized confidence is poorly calibrated. Yet no work has directly compared these signals on the same benchmark with the same model, leaving practitioners without guidance on which approach to adopt.

The gap we address: Prior work assumes that combining uncertainty signals should help, but this assumption has not been tested. We provide the first systematic comparison of entropy and consistency signals on identical evaluation conditions, and we test whether their combination improves detection. The answer—at least at proof-of-concept scale—is no.

1.2 Key Insight

Our central finding is that **semantic consistency alone is sufficient** for hallucination detection in factual QA, achieving strong discrimination (AUROC=0.81) without requiring entropy computation. This is practically significant: consistency requires only generated text, while entropy requires logit access—excluding API-only models like GPT-4 and Claude. More fundamentally, the failure of linear fusion to improve over consistency alone challenges the intuition that “more signals are better.”

We interpret this result through the lens of signal redundancy. The moderate negative correlation ($r=-0.54$) between entropy and consistency means they partially overlap: questions where the model has high entropy (internal uncertainty) tend to produce low-consistency outputs (unstable responses). This correlation reduces the information gain from combining them.

However, we emphasize an important limitation: our fusion experiments used only $N=20$ questions due to computational constraints. The grid search for fusion weights may have overfit to this small sample, collapsing to pure consistency (optimal weights: $\alpha = 0, \beta = 0.1$). The complementarity hypothesis is **inconclusive**, not refuted—larger-scale validation is needed.

1.3 Contributions

This paper makes three contributions:

1. **First direct AUROC comparison of entropy and consistency signals** on the same benchmark (TriviaQA) with the same model (Llama-2-7B-chat) under identical evaluation conditions. We find consistency outperforms entropy by 16 percentage points (0.81 versus 0.65).
2. **Quantified effect sizes** demonstrating that consistency is not merely statistically significant but practically meaningful: Cohen’s $d=1.19$ (large effect) versus $d=0.47$ (moderate) for entropy. This provides actionable guidance for practitioners.
3. **An important negative result:** Linear fusion of entropy and consistency does not improve detection at proof-of-concept scale. This challenges the assumption that combining uncertainty signals provides additive benefit and identifies the entropy-consistency correlation ($r=-0.54$) as a limiting factor.

Our results suggest that practitioners working on hallucination detection in factual QA should prioritize consistency-based methods. The additional complexity of computing token entropy—which requires logit access and thus excludes commercial API-only models—does not appear justified by improved detection, at least in the regime we tested.

2 Related Work

Uncertainty quantification for large language models has developed along several parallel tracks. We organize prior work into three categories—entropy-based methods, consistency-based methods, and verbalized confidence—then position our contribution relative to this landscape.

2.1 Entropy-Based Uncertainty Estimation

Token-level entropy, computed from the softmax distribution over next-token predictions, has long served as a proxy for model uncertainty in language models. High entropy indicates a flat distribution where the model assigns similar probability to many continuations, signaling low confidence.

Kuhn et al. [2023] extended this approach with **semantic entropy**, which clusters generated responses by meaning before computing entropy. Their key insight is that multiple *phrasings* of the same answer should not inflate uncertainty—only semantically distinct responses matter. On closed-book QA benchmarks, semantic entropy achieves AUROC in the range 0.75–0.85, substantially outperforming raw token entropy.

However, entropy-based methods require access to model logits, excluding their application to commercial API-only models (GPT-4, Claude). They also conflate different sources of uncertainty: a model might be uncertain about word choice (low semantic impact) or uncertain about factual content (high semantic impact).

Our work builds on entropy-based methods by including token entropy as a baseline signal, but we find it underperforms consistency even without semantic clustering—achieving only AUROC=0.65 on TriviaQA with Llama-2-7B-chat.

2.2 Consistency-Based Hallucination Detection

Consistency-based methods leverage the intuition that models produce stable outputs for questions they can reliably answer, but divergent outputs when hallucinating.

Manakul et al. [2023] introduced **SelfCheckGPT**, which generates multiple responses and measures their consistency via BERTScore, question-answering overlap, or n-gram similarity. When responses contradict each other, the model is likely hallucinating. They demonstrated strong performance on WikiBio-generated biographies but did not report AUROC on standard QA benchmarks like TriviaQA.

Wang et al. [2023] explored **self-consistency** in the context of chain-of-thought reasoning, finding that majority voting over multiple reasoning paths improves accuracy. While their focus was on improving outputs rather than detecting errors, the underlying signal—consistency across samples—is the same.

Our approach follows the SelfCheckGPT paradigm but uses embedding cosine similarity (via Sentence-Transformer all-MiniLM-L6-v2) rather than BERTScore. We provide the first AUROC quantification of this signal on TriviaQA, finding AUROC=0.81 with a large effect size ($d=1.19$).

2.3 Verbalized Confidence

An alternative approach asks models to state their own confidence explicitly. Xiong et al. [2023] studied **verbalized confidence** across multiple benchmarks, prompting models to rate their certainty. They found that verbalized confidence is poorly calibrated—models often express high confidence in wrong answers.

Lin et al. [2022] similarly found that models’ self-reported uncertainty does not correlate well with actual correctness, particularly for instruction-tuned models trained to sound confident.

We do not include verbalized confidence as a baseline because it requires prompt engineering and exhibits high variance across phrasings. Our focus is on signal-based methods that derive uncertainty from model behavior rather than self-report.

2.4 Positioning Our Contribution

Prior work has developed entropy and consistency signals largely in isolation. Kuhn et al. [2023] focused on improving entropy via semantic clustering; Manakul et al. [2023] focused on improving consistency via alternative similarity metrics. Neither directly compared the signals on the same benchmark under controlled conditions.

What prior work leaves unanswered:

- Which signal—entropy or consistency—is more predictive for factual QA?
- Do the signals capture complementary information such that combining them improves detection?
- What is the correlation between these signals, and does it limit fusion benefit?

We address these questions by evaluating both signals on TriviaQA with Llama-2-7B-chat under identical conditions. Our contribution is not a new method but a systematic comparison that provides actionable guidance.

3 Methodology

We design a controlled experiment to compare token entropy and semantic consistency as hallucination predictors, then test whether their combination improves detection.

3.1 Dataset

We use **TriviaQA** [Joshi et al., 2017], specifically the `rc.nocontext` split from HuggingFace. This configuration presents questions without accompanying documents, testing the model’s parametric knowledge rather than reading comprehension. TriviaQA provides ground-truth answers for each question, enabling automatic evaluation of response correctness.

Correctness labeling: A response is marked correct if it achieves either (1) exact match with any reference answer after normalization, or (2) token-level F1 > 0.5 with the reference.

3.2 Model

We use **Llama-2-7B-chat** [Touvron et al., 2023], an instruction-tuned variant of the Llama-2 7B base model. This model represents a widely-deployed class of open-source LLMs with full logit access, enabling entropy computation.

3.3 Response Generation

For each question, we generate **10 independent responses** using nucleus sampling with temperature 0.7, top-p 0.95, and max new tokens 128. We use a fixed random seed (42) for reproducibility.

3.4 Token Entropy Computation

We compute **Shannon entropy** from the token-level softmax distributions:

$$H(y_t) = - \sum_{v \in V} p(v|y_{<t}) \log p(v|y_{<t}) \quad (1)$$

We aggregate by taking the mean across all generated tokens in the response, then average across all 10 responses to obtain a single entropy score per question.

3.5 Semantic Consistency Computation

We measure consistency via **pairwise embedding similarity** across the 10 generated responses:

1. Encode each response using SentenceTransformer (`all-MiniLM-L6-v2`)
2. Compute cosine similarity between all 45 pairs
3. Average pairwise similarities to obtain the consistency score

Higher consistency indicates stable model behavior; lower consistency indicates hallucination-prone responses.

3.6 Linear Fusion

To test combination, we apply **linear fusion**:

$$\text{Score} = \alpha \cdot \text{Confidence} + \beta \cdot \text{Consistency} \quad (2)$$

where $\text{Confidence} = 1 - \text{normalized_entropy}$. Weights α, β are determined via grid search over $[0, 1]$ with step 0.1 using a held-out validation split.

3.7 Evaluation Metrics

- **AUROC:** Area under ROC curve for correctness prediction
- **Cohen’s d:** Effect size between correct/incorrect distributions
- **Statistical significance:** Two-sample t-test with $\alpha = 0.05$
- **Pearson correlation:** Between entropy and consistency scores

4 Experiments

4.1 Research Questions

RQ1 (Feasibility): Can both metrics be reliably computed?

RQ2 (Entropy Signal): Does higher entropy correlate with incorrect answers?

RQ3 (Consistency Signal): Does lower consistency correlate with incorrect answers?

RQ4 (Fusion Benefit): Does combining signals improve over the best single metric?

4.2 Hypotheses and Gates

Hypothesis	Gate	Criteria
H-E1: Metrics computable	MUST_WORK	>99% success rate
H-M1: Entropy correlates	MUST_WORK	$p < 0.05$, AUROC > 0.55
H-M2: Consistency correlates	SHOULD_WORK	$p < 0.05$, AUROC > 0.55
H-M3: Fusion improves	SHOULD_WORK	AUROC_combined > max(single)

4.3 Sample Sizes

H-M1 uses $N=100$ questions for statistical power. H-M2 and H-M3 use $N=20$ questions as proof-of-concept, acknowledged as a limitation.

5 Results

5.1 H-E1: Metric Computability

Both metrics achieved 100% computation success. **Gate: PASS.**

Metric	Mean	Std	Range
Entropy	0.1224	0.0395	[0.04, 0.25]
Consistency	0.8693	0.0748	[0.65, 0.98]

5.2 H-M1: Entropy-Correctness Correlation ($N=100$)

Metric	Correct (n=60)	Incorrect (n=40)
Mean entropy	0.1104	0.1342

Gate: PASS

Test	Value	Status
p-value	0.0246	PASS
AUROC	0.6454	PASS
Cohen’s d	0.47	Medium effect

5.3 H-M2: Consistency-Correctness Correlation (N=20)

Metric	Correct (n=11)	Incorrect (n=9)
Mean consistency	0.896	0.826

Test	Value	Status
p-value	0.0083	PASS
AUROC	0.808	PASS
Cohen’s d	1.19	Large effect

Entropy-Consistency Correlation: Pearson $r = -0.54$
Gate: PASS

5.4 H-M3: Linear Fusion (N=20)

Improvement: 0.000 — Fusion collapsed to pure consistency.
Gate: FAIL

5.5 Summary

Key finding: Consistency outperforms entropy by 16 percentage points; fusion provides no additional benefit at PoC scale.

6 Discussion

6.1 Why Consistency Outperforms Entropy

Consistency measures behavioral outcomes (do multiple attempts agree?) while entropy measures internal states (how uncertain is the token distribution?). When a model hallucinates, it may generate confident but inconsistent responses across samples. Entropy fails to flag these cases; consistency captures them.

The large effect size for consistency ($d=1.19$ vs $d=0.47$) indicates this is practically meaningful, not merely a statistical artifact.

6.2 Why Fusion Failed

Three factors explain the collapse to pure consistency:

1. **Small sample size:** $N=20$ with 2-sample validation cannot reliably estimate fusion weights
2. **Consistency dominance:** The stronger signal (0.81) overwhelms the weaker (0.65)
3. **Linear assumption:** Non-linear combinations might extract value that linear fusion misses

We emphasize that H-M3’s failure does not disprove fusion’s potential—it demonstrates that at PoC scale, consistency alone suffices.

Variant	Alpha	Beta	AUROC
Entropy only	1.0	0.0	0.675
Consistency only	0.0	1.0	0.812
Optimal	0.0	0.1	0.812

Hypothesis	AUROC	Cohen’s d	Verdict
H-M1 (Entropy)	0.645	0.47	PASS
H-M2 (Consistency)	0.808	1.19	PASS
H-M3 (Fusion)	0.812	—	FAIL

6.3 Limitations

- **Sample size:** N=20 for H-M2/H-M3 limits statistical power
- **Single model:** Llama-2-7B-chat only; larger models may differ
- **Single dataset:** TriviaQA only; other tasks may show different patterns
- **Entropy baseline:** We compare against raw token entropy, not semantic entropy [Kuhn et al., 2023], which clusters responses by meaning before computing entropy and achieves higher AUROC (0.75–0.85). A direct comparison with semantic entropy remains future work.
- **Inference cost:** Consistency requires 10 forward passes per question (for 10 samples), while entropy can be computed from a single pass. This 10x cost difference is critical for practitioners weighing deployment tradeoffs.

6.4 Practical Implications

1. **Start with consistency** if you can afford multiple generations per query
2. **Entropy as fallback** when multiple generations are impractical
3. **Skip fusion at small scale** until validated on your specific task

7 Conclusion

We presented the first systematic comparison of token entropy and semantic consistency for hallucination detection. Our key findings:

1. **Consistency dramatically outperforms entropy:** AUROC 0.81 versus 0.65, Cohen’s d 1.19 versus 0.47
2. **Linear fusion provides no improvement** at proof-of-concept scale (N=20)
3. **Signals are moderately correlated** ($r=-0.54$), suggesting partial redundancy

These results challenge the assumption that combining uncertainty signals automatically improves detection. For practitioners building hallucination detection systems, consistency should be the foundation. The complementarity hypothesis requires larger-scale validation before practical deployment.

Future work should validate on larger samples ($N \geq 500$), extend to additional benchmarks (Natural Questions, TruthfulQA), explore non-linear combination strategies, and test across model scales (13B, 70B).

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